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Mechanism of convection-dependence of leveller and its effect on copper pillar bump filling uniformity

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ABSTRACT

Metallisation processes, represented by copper pillar bumps (CPBs), are taking the spotlight in the field of semiconductor advanced packaging. Efficient filling of a uniform, levelling copper layer within a defined pattern by electroplating is the final goal pursued. In addition to its levelling capabilities, the leveller Janus Green B (JGB) has an undesired convection-dependence for CPBs, resulting in a convex front upstream or downstream of the copper filling layer. In this paper, the JGB is identified as the source of convection-dependence based on an *in-situ* observation platform. Electrochemical and spectroscopic evidence supports the conclusion that the reduction reaction (including protonation of the phenazine ring, hydrogenation reduction or even further cleavage of the azo group) of JGB also occurs along with the copper electrodeposition process. The introduction of convection produces a stronger suppression effect and thus a non-uniform distribution of the filled layer within the microcavity.

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KEYWORDS
Copper pillar bumps; filling uniformity; convection-dependence; leveller; Janus Green B

1. Introduction

The deposit leveller takes an important role in plating-related processes such as copper damascene interconnection,[1–3] copper pillar bumps,[4,5] and microvias.[6,7] Janus Green B (JGB) is the commonly used leveller which has been studied extensively. Kelly *et al.*[1] and Cao *et al.*[8] firstly proposed a four-component additive containing JGB to achieve the levelling effect and suggested a competitive adsorption relationship between these additives. Dow *et al.*[9–11] found that the inhibiting effect of JGB was enhanced in the presence of convection and proposed the convection-dependent adsorption (CDA) theory, suggesting that the CDA behaviour of the additive was a crucial factor to achieve bottom-up filling. Li *et al.*[12] determined the structure of the reduction product of JGB by mass spectrometry (MS) and nuclear magnetic resonance (NMR), and proposed that the reduction product of JGB is the true leveller in the copper filling process. Mao *et al.*[13] provided molecular spectroscopic evidence on the reductive decomposition of the JGB azo group by *in-situ* attenuated total reflection-surface enhanced infrared spectroscopy. Hai *et al.*[14] further stated that the consumption of JGB should also include the reduction of heterocycles. In the present authors' previous work, JGB-induced convection-dependence has been observed,[15] but molecular-level understanding of its genesis is still absent.

In this study, the JGB-induced convection-dependent filling process is visualised as benefitting from the CPBs *in-situ* observation platform. Cyclic voltammetry (CV), ultra violet-visible spectroscopy (UV-vis), and *in-situ* electrochemical surface enhanced Raman spectroscopy (EC-SERS) were performed to identify the decomposition of JGB. The weakening of the inhibitory capacity of JGB reduction product relative to JGB was confirmed by cyclic voltammetry stripping (CVS). Electrochemical and spectroscopic evidence supports the idea that inhomogeneous flow field inside the microvia leads to enhanced renewal of JGB and its reduction product on the higher flow rate side, and suppressed the deposition of copper on that side.

2. Experimental

2.1. Materials

H_2SO_4, NaCl, and polyethylene glycol (PEG, mol. wt. 6000) were purchased from Sinopharm Chemical Reagent Co. Ltd. (China). $CuSO_4{\cdot}5H_2O$, sodium 3,3′-disulfanediylbis(propane-1-sulphonate) (SPS) and methylene violet 3RAX (MEV) was obtained from Adamas (China). Merck (Germany) supplied JGB.

2.2. Set-up of the in-situ observation platform

A microcavity electrode (MCE) (aspect ratio, AR ~1) was etched and assembled in an *in-situ* observation platform according to the authors' previous work.[15] In brief, epoxy-sealed nickel wire was etched in 1 M NaCl + 5 mM H_2SO_4 to form the MCE, which was then placed in the trench ($5 \times 20 \times 5$ mm) of the Polyetheretherketone (PEEK) cell. A 1 mm platinum wire and a 0.7 mm copper wire embedded in the cell were used as counter and reference electrode, respectively. The cell was sealed by an epoxy-coated 0.5 mm thickness glass cover plate, and the plating bath was circulated through the fluidic chamber ($30 \times 5 \times 7$ mm) at a rate of 1.26 cm s^{-1} by means of a peristaltic pump (Kamoer, DI Pump 550, China).

2.3. Electrochemical experiments

Cyclic voltammetry was performed to evaluate the electrochemical behaviour of JGB with 3 mm glassy carbon (GC) electrode as the working electrode, graphite rod and Hg/Hg_2SO_4 (sat.) (MSE) as the counter electrode and reference

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electrode, respectively. The electrolysis of JGB was performed by applying 6 A dm^{-2} (ASD) for 1 h between the copper sheet cathode and the platinum sheet anode in the double chamber electrolytic cell separated by a Nafion 117 proton membrane (DuPont, USA). The basic electrolyte solution (BES) was composed of 5 ppm JGB, 60 ppm Cl^- and 0.5 M H_2SO_4. An equivalent concentration of 0.5 ppm of JGB and its reduction products obtained from the 1 h electrolysis experiment (diluted 10-fold) were added to the basic bath, respectively. The cyclic voltammetry stripping was performed with a potential window of −0.4–1.0 V (*vs*. $Cu/CuSO_4$ (sat.), CCSE) at a scan rate of 100 mV s^{-1}. The inhibition factor was defined as the ratio of the area of the Cu stripping peaks after and before additive injection, and was obtained by three independent experiments. All electrochemical experiments were performed by the CHI 660D Potentiostat (CH Instrument, China) at room temperature.

2.4. EC-SERS measurement

The signals of Raman spectra were collected by a Raman spectrometer (Renishaw, Invia, UK) equipped with 632.8 nm He–Ne laser and 50× long focus objective. The preparation of $Au@SiO_2$ (~55 nm Au core with ~2 nm silica shell) nanoparticles (NPs) was as given in the previous report by Li *et al*.[16] The prepared $Au@SiO_2$ NPs were drop-coated onto the surface of a 3 mm GC electrode and dried under vacuum. Platinum wire and MSE were used as counter electrode and reference electrode, respectively. The potentials were stepped in the negative direction from open circuit potential (OCP, 0.015 V *vs* MSE) to −0.7 V at intervals of 0.1 V in BES, and held constant for 50 s at each potential for Raman spectra acquisition.

2.5. Characterisations

The copper filling process was recorded in an optical microscope (SANQTID, China). The surface morphology was characterised by scanning electron microscopy (SEM, Quanta 250, FEI, USA). The UV-vis spectra were obtained by a double-beam spectrophotometer (TU-1950, Persee, China). Transmission electron microscopy (TEM-HT7700, Hitachi, Japan) was used for characterisation of $Au@SiO_2$ NPs.

3. Results and discussion

3.1. Convection-dependent filling layer profiles due to leveller JGB

As shown in Figure 1, three different convex fronts were captured during the filling of the MCEs. The combination of PEG with SPS produced a central convexity, which was caused by localised adsorption and accumulation of reactant. The introduction of JGB was originally intended to flatten the surface, but created a downstream bulge problem. Furthermore, an unexpected contrast is the upstream bias front when only JGB is added.

Figure 2 shows the surface morphology of MCEs obtained under different additive combinations. While the addition of PEG and SPS results in refined grains with respect to the basic plating bath, the lower roughness surface clearly requires the injection of JGB. Therefore, taking into account the roles played by the flow field and the leveller JGB in the convection-dependent morphology generation, it is necessary to reconsider the properties of JGB.

3.2. Electrochemical behaviour of JGB

To understand how JGB produces convection-dependent effect during deposition, the authors first studied the electrochemical behaviour of JGB as shown in Figure 3. The Glass carbon was chosen as the work electrode due to its wider anti-interference potentials window and easier identification of peaks associated with JGB.

Figure 3(a) shows five different observed peaks when CV scans were performed with different concentrations of JGB in an electrolyte consisting of 0.5 M H_2SO_4 and 60 ppm Cl^-. Peaks 1, 2, 3 and 5 exhibit concentration dependence with JGB and nearly approach saturation at concentrations above 800 ppb.

Figure 3(b) illustrates the unexpected electrochemical behaviour of JGB found during the multi-cycle CV scans. The JGB-related peaks described above almost completely

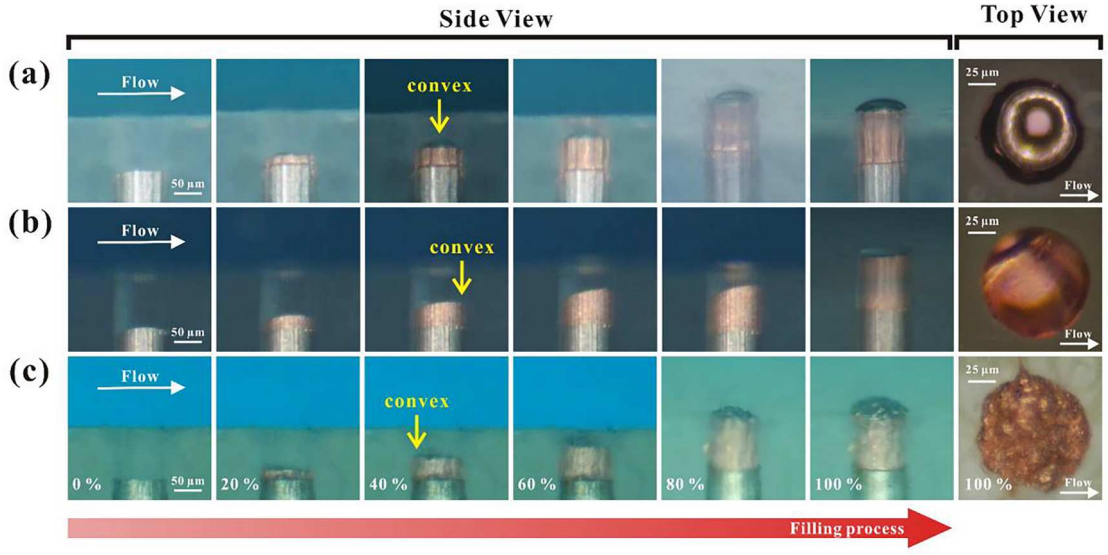


Figure 1. Copper filling process of 100 μm MCE (AR ~1) with bath composed of: (a) Basic + 500 ppm PEG + 5 ppm SPS, (b) Basic + 500 ppm PEG + 5 ppm SPS + 5 ppm JGB, and (c) Basic + 5 ppm JGB. Basic: 1 M $CuSO_4$ + 0.5 M H_2SO_4 + 60 ppm Cl^-.

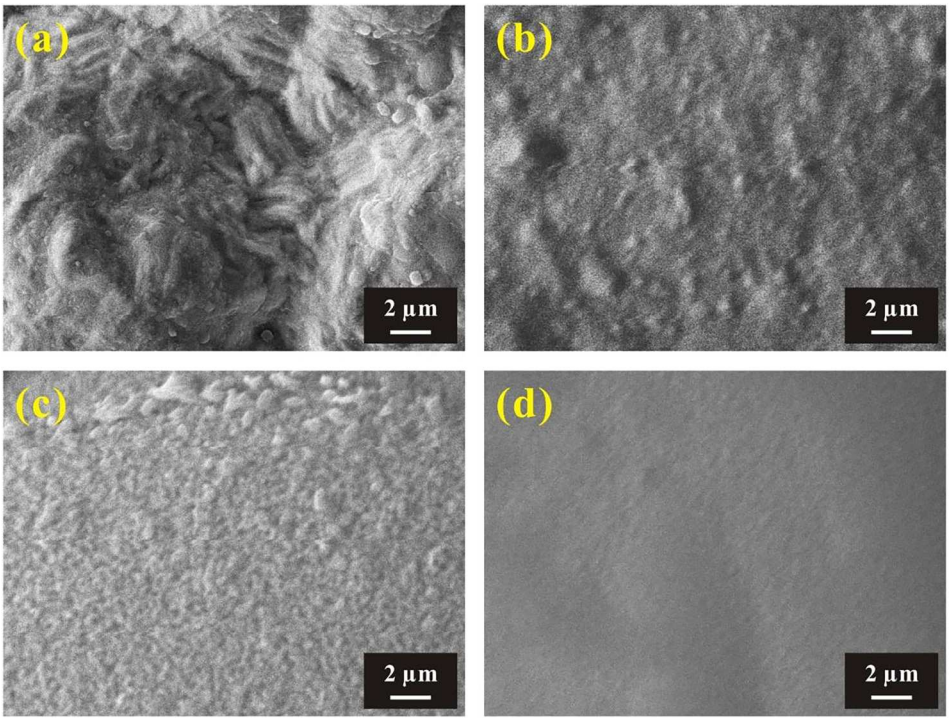


Figure 2. Surface morphology of filled MCEs with bath composed of: (a) Basic, (b) Basic + 500 ppm PEG, (c) Basic + 500 ppm PEG + 5 ppm SPS, and (d) Basic + 500 ppm PEG + 5 ppm SPS + 5 ppm JGB. Basic: 1 M $CuSO_4$ + 0.5 M H_2SO_4 + 60 ppm Cl^-.

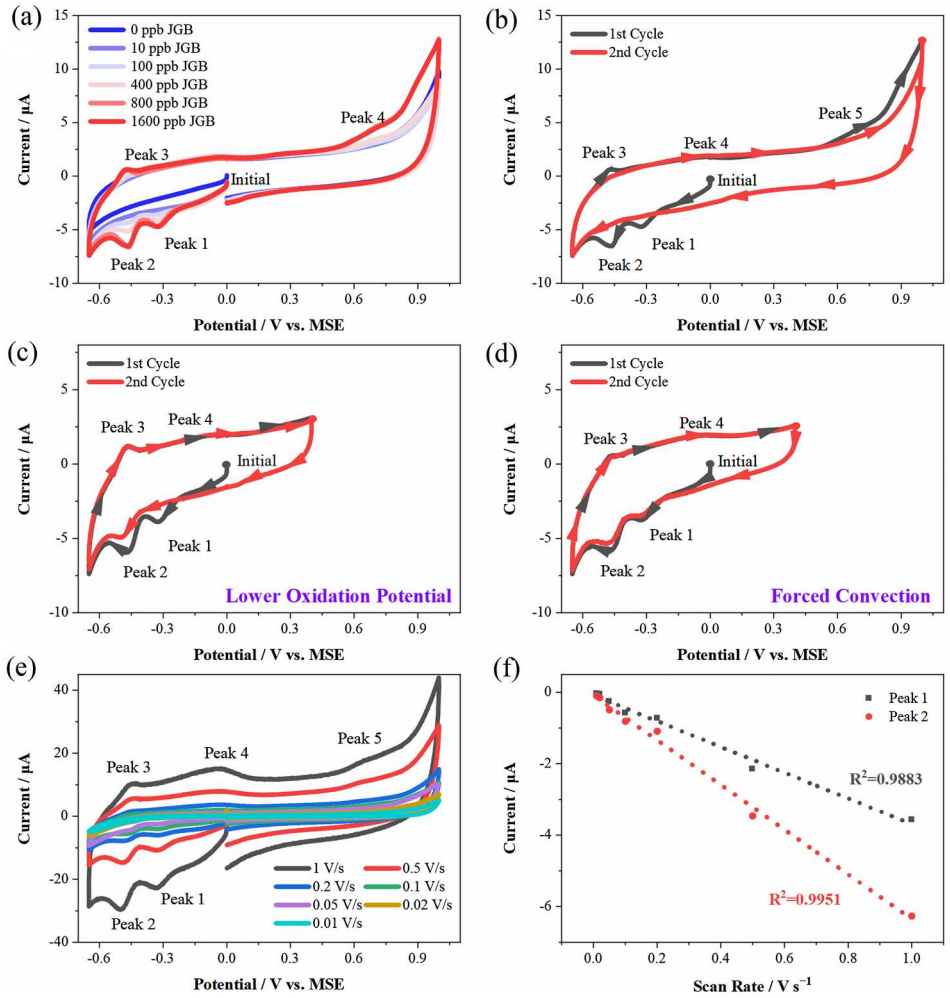


Figure 3. (a) Electrochemical behaviour of JGB at different concentrations in solutions containing 0.5 M H_2SO_4 and 60 ppm Cl^- on the surface of Φ3 mm GC electrode, scan rate: 100 mV s^{-1}. (b) The redox peaks associated with JGB all disappeared at the second cycle (red line) in two consecutive CV scannings with solution consisting of 0.5 M H_2SO_4, 60 ppm Cl^- and 1600 ppb JGB, where (c) peaks 2 and 3 can be recaptured after lowering the oxidation potential, and (d) peak 1 needs to be observed again after increasing forced convection (magnetic stirring at 600 rpm). (e) Variation of all peak currents with scan rate (0.01–1 V s^{-1}). (f) Relationship between scan rate and the reduction current peaks of interest (peak 1 and peak 2) as shown in (e).

disappeared in the second cycle. And when lowering the positive reversal potential to 0.4 V (Figure 3(c)), peaks 2 and 3 reappeared, while peak 1 remained absent, suggesting that irreversible reactant would be generated at higher potential. Peak 1 in the second cycle did not reappear until the introduction of forced convection (Figure 3(d)), which reminds us of the convection-dependence of JGB.

The effect of scan rate on the peak current was further examined (Figure 3(e)), and the relationship between scan rate and peak current of interest is shown in Figure 3(f). The peak currents of peak 1 and peak 2 are linearly correlated with the scan rate, which implies that the reaction rates of both are mainly controlled by adsorption.

It can be inferred from the above experiments that JGB undergoes the first electron transfer at −0.32 V, followed by another reduction at −0.46 V. The reduced species will form a hydrophobic 3D cluster precipitation product on the surface similar to that observed by Hai *et al.*,[14] which may arise from cationic polymerisation reactions during electrochemical processes.[17] The reduced species do not leave the electrode surface directly, while the introduction of convection can accelerate the exchange between JGB and its redox, thus exposing new sites for adsorption.

Taking into account that the deposition of copper has more negative reduction potential than the reduction of JGB, it is suggested that the reduction of JGB during the plating process generates a new product (JGB_Red), which plays a real role in surface inhibition.[12,14,18] The flow rate difference within the microvias leads to an accelerated exchange rate of JGB-JGB_Red on the surface, thus exhibiting different deposition rates on the left and right sides of microvias.

3.3. Spectroscopic study of JGB

To further confirm the consumption of JGB, an electrolysis experiment was performed with an H-type electrolytic cell separated by a proton membrane. The solutions containing the reduced and oxidised products of JGB were obtained respectively as shown in Figure 4(a). The solution of the cathode chamber changed from blue to violet colour, while the blue colour of the anode chamber gradually faded.

Figure 4(b) shows the UV-vis spectra of JGB and its redox products in neutral and acidic solutions. The aqueous solution of JGB exhibits absorption peaks at 284, 393, and 603 nm. As a comparison, the maximum absorption peak at 603 nm is significantly broadened in the acidic solution, and two new peaks at 557 and 616 nm appear. Considering that the absorbance at 450–750 nm corresponds to the $n\rightarrow\pi^*$ transitions (involving electron transitions from lone pairs of the nitrogen atoms), this difference is supposed to be the new transitions and the coupling with other transitions due to the JGB protonation in acidic aqueous solution.

Figure 4(b) also implies the generation of species distinct from JGB after electrolysis, especially the presence of maximum absorption peaks at 377 and 557 nm for the reduction product of JGB (red line). The authors speculate that this is a further hydrogenation reduction of the azo group (disappearance of the absorption band at 405 nm correlated with the azo group).[19] In contrast, a clear decrease in the concentration of JGB appears in the anodic compartment (weakened absorbance).

For comparison, the UV-vis spectra of MEV, which is structurally similar to the phenazine ring of JGB, were also collected in Figure 4(b). The characteristic absorption peaks of MEV at 377 and 557 nm correspond exactly to the JGB reduction product. This result confirms that the electrochemical reduction of JGB will lead to the cleavage of the azo group. A new peak at 377 nm was also observed for MEV in acidic solution, corresponding to the protonation of phenazine.

Subsequently, the decomposition process of JGB was further investigated by EC-SERS experiments. Based on the Shell isolated nanoparticles enhanced Raman spectroscopy (SHIENRS) technique, the Au@SiO_2 NPs are distributed on the surface of the GC electrode to achieve signal

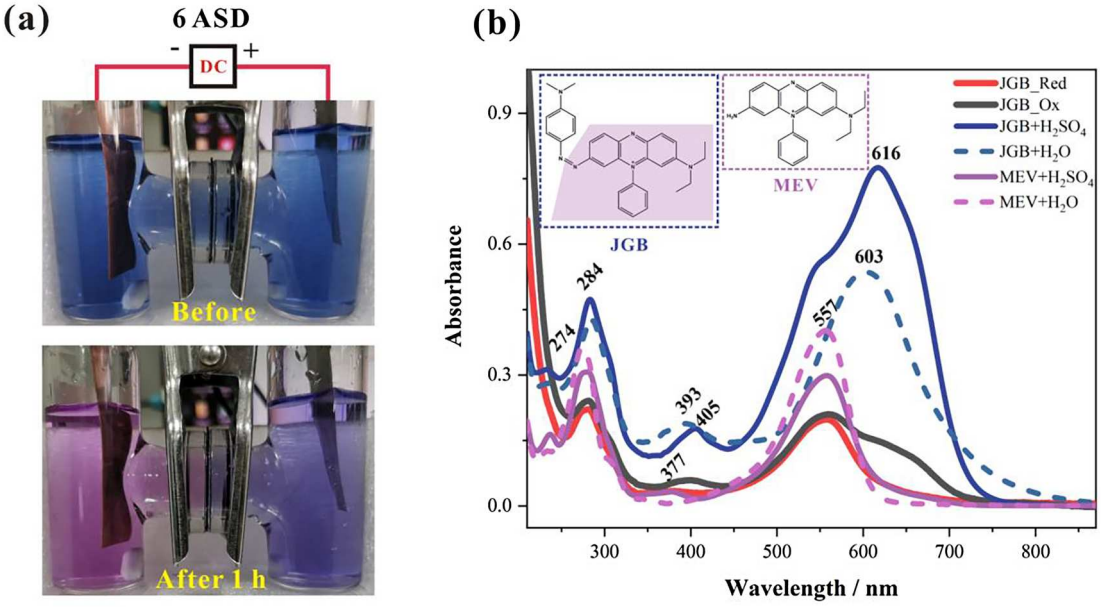


Figure 4. (a) Pictures of cathode and anode chambers before (top) and after (bottom) 1 h of electrolysis at 6 ASD in the solution composed of 0.5 M H_2SO_4, 60 ppm Cl^- and 5 ppm JGB. (b) UV-vis spectra of MEV (purple line) and JGB (blue line) in 0.5 M H_2SO_4 (solid line) and water (dashed line), JGB reduction product (red line), oxidation product (grey line) after 1 h of electrolysis

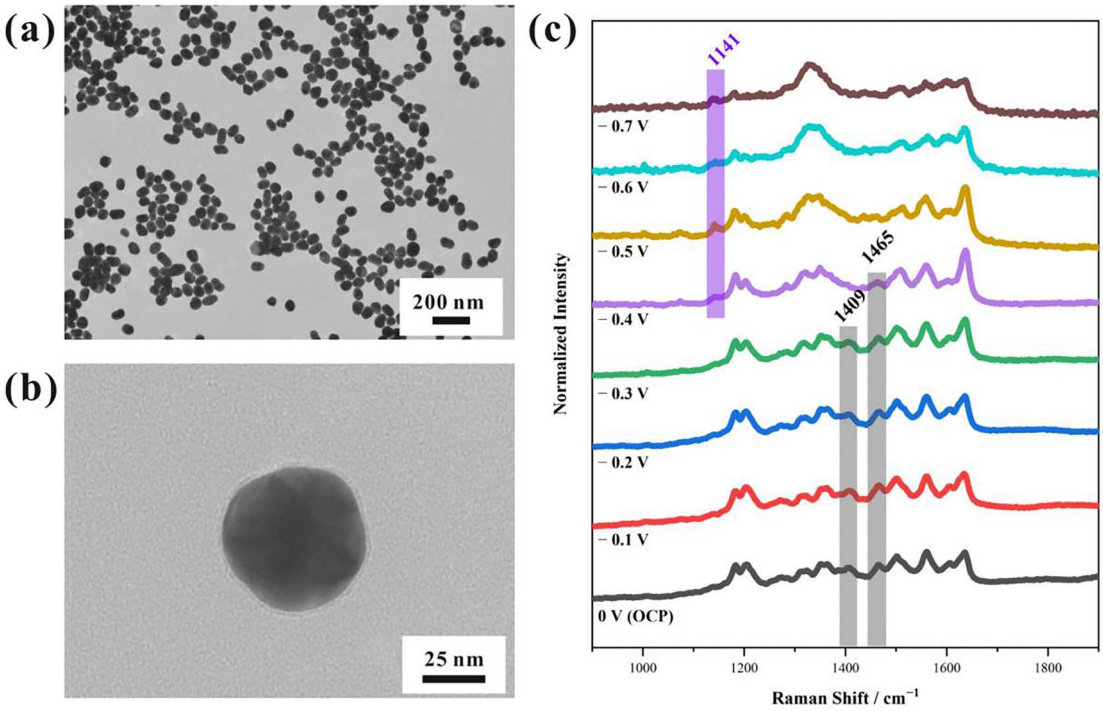


Figure 5. (a) Characterisation of Au@SiO_2 NPs. (b) HR-TEM images of Au@SiO_2 (~55 nm Au core with ~2 nm silica shell) to visualise clear profiles of Au core and SiO_2 shell. (c) EC-SERS spectra of JGB on the surface of GC electrode coated with Au@SiO_2 NPs in BES.

enhancement by the plasmonic resonant excitation of the gold nucleus, while the silica shell avoids the interference of environmental factors on the signal.[16,20,21]

Figure 5(a,b) characterises the profile morphology of Au@SiO_2 NPs (55 nm Au, 2 nm SiO_2) synthesised according to the previously reported method. The EC-SERS signals obtained by coating the Au@SiO_2 NPs on the surface of the GC electrode in BES containing 5 ppm JGB are shown in Figure 5(c).

As shown in Figure 5(c), with the negative shift of the potential, the stretching vibrational modes of the azo group at 1409, 1465 cm^{-1} were observed to gradually weaken, and disappear completely at – 0.4 V.[21,22] At the same time, a new vibrational mode appeared at 1141 cm^{-1}, which corresponded to the stretching vibration of the N-N single bond, suggesting that the azo group underwent a hydro reduction reaction.[21,23] Although the N-H vibration modes at higher wavelengths are masked by strong fluorescence signals and are difficult to recognise by Raman spectroscopy, recently reported infrared spectroscopy experiments have confirmed that the reduction of the JGB is even accompanied by the complete breakage of the azo group.[13]

Despite the fact that the identification of the specific structure of the JGB reduction product is not the focus of this section of the study, a combination of UV-vis spectroscopy and *in situ* electrochemical Raman spectroscopy experiments has allowed the following conclusions to be drawn: The reduction reaction of JGB occurred at the copper deposition potential window. These reduction processes involve protonation of the phenazine ring,[17,24–26] hydrogenation reduction or even further cleavage of the azo group.[12–14,21] The resulting reduced species may act as the real leveller. This electrochemical reduction property of JGB is hypothesised to be responsible for its ability to produce convection-dependent behaviour.

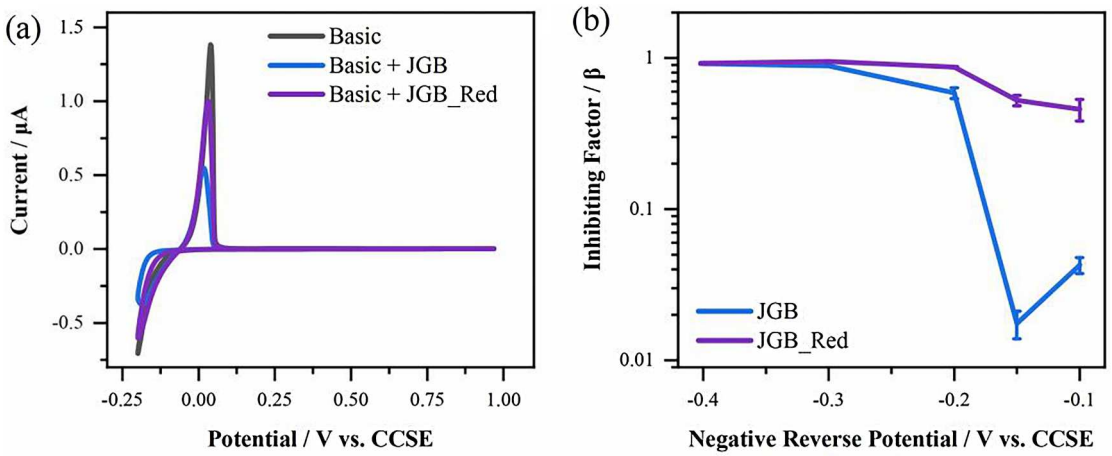


Figure 6. (a) CVS curves in plating bath without (grey) and with 0.5 ppm JGB (blue) or JGB_Red (purple). (b) Trend of inhibiting factors of JGB and JGB_Red with different negative reverse potentials. Basic: 1 M $CuSO_4$ + 0.5 M H_2SO_4 + 60 ppm Cl^- .

3.4. Mechanism of convection-dependence of leveller

As a final component of this study, the difference in inhibition of copper deposition by JGB and JGB_Red needs to be evaluated. The cyclic voltammetry stripping (CVS) method[27] was used to compare the changes in the inhibition ability of the plating solution when JGB or JGB_Red was added to the basic plating bath (Figure 6).

Figure 6(a) illustrates typical CVS results used to evaluate additive inhibition capability. Additives with inhibition ability will slow down the deposition rate of Cu, which further leads to a decrease in the area of the Cu dissolution peak at around 0 V.

Figure 6(b) presents the trend of JGB and JGBRP inhibition factors with potential. The negative reverse potential was varied from −0.1 to −0.4 V to cover the potential window for JGB reduction as well as copper deposition. As shown in Figure 6(b), both JGB and JGBRP exhibited the ability to inhibit copper deposition. However, when the potential is shifted negatively to −0.3 V, the inhibition ability of both decreases significantly, which is attributed to the desorption of the leveller at more negative potentials.

JGB exhibits a consistently stronger inhibition ability than its reduction product, indicating that the inhibition ability of JGB can be maintained at a high level even if it decomposes when the raw material is sufficiently supplied. Therefore, even with the generation of JGB_Red in actual plating, the weakened inhibition ability of JGB can be compensated for as long as sufficient convection is ensured.

So far, based on the above spectroscopic evidence and electrochemical studies, the convective dependence exhibited by JGB can be explained: JGB initially protonates in acidic solutions and subsequently becomes further decomposed during copper deposition, and the inhibition of the redox is much weaker. For regions with strong convection (e.g. mouth of the microvia,[9,10] and downstream of microcavity[15,28]), where sufficient raw material supply can maintain the inhibition ability, the growth of copper can be slowed down, while, for the weaker convective regions (e.g. bottom of the microvia, and upstream of microcavity), the inhibition is relatively weak, which will be preferred for copper growth.

4. Conclusion

In this paper, the mechanism of convection-dependent morphology induced by the leveller JGB has been investigated based on an *in-situ* observation platform simulating the work conditions of CPBs electrodeposition. The results of CV, UV-vis, and EC-SERS experiments indicate that the decomposition of JGB involves the protonation of the phenazine ring and the reduction of the azo group. The CVS studies further indicate that the reduction of JGB will result in a weakening of its inhibitory capacity. Enhanced convection can ensure sufficient supply of JGB to maintain its inhibitory capacity, while the decomposition of JGB in the weak convection region will deteriorate the capacity. This is believed to be the reason why the plating layer height distribution is affected by convection during the filling process. The revelation of this decomposition mechanism will contribute to the understanding of the role played by phenazine-type leveller, and facilitate the molecular design of leveller and the quality control of plating bath in the CPBs and microvia filling process.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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